

JINWOO KIM

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Research Interest

My focus is in making deep learning models generalize better beyond their training data so that they can be used to solve challenging problems, such as those in scientific domains. I am studying this problem from the complementary perspectives of invariance and equivariance, parallel neural processing, and Solomonoff induction. I often use tools from theories of graphs, manifolds, (semi)groups and categories, as well as Markov processes such as random walks, diffusions, and flows.

Academic Placement and Employment

TU Wien , Visiting Scholar (host: İsmail İlkan Ceylan)	(planned) 06/2026 – 09/2026
KAIST , Technical Research Personnel (host: Seunghoon Hong)	03/2026 – 02/2027
New York University , Visiting Scholar (host: Kyunghyun Cho, Rajesh Ranganath)	11/2025 – 01/2026
LG AI Research , Research Intern (host: Moontae Lee, Honglak Lee)	01/2022 – 07/2022

Education

KAIST , M.S./Ph.D. in Computer Science (advisor: Seunghoon Hong)	03/2021 – 02/2026
- Ph.D. Thesis: Architecture-Agnostic Invariances for Deep Learning	
KAIST CoE Best Ph.D. Dissertation Award	
KAIST , B.S. in Brain Engineering and Computer Science (double major)	03/2016 – 02/2021
- GPA 4.05/4.3 (Summa Cum Laude)	

Publications

C: conference, J: journal, W: workshop, P: preprint, *: equal contribution, †: equal advising

P5. **Synthetic Data, Compression, and Generalizable Representations**

Jinwoo Kim, Jiyun Park, Kyunghyun Min, Hoyeon Chang, Seunghoon Hong, Kyunghyun Cho
In preparation

P4. **What are the Right Symmetries for Formal Theorem Proving?**

Krzysztof Olejniczak, Radoslav Dimitrov, Xingyue Huang, Bernardo Cuenca Grau, Jinwoo Kim[†], İsmail İlkan Ceylan[†]
arXiv 2026

P3. **MUX: Continuous Reasoning via Multiplexed Tokens**

Ayhan Suleymanzade, Halil Alperen Gozeten, Michael Bronstein, İsmail İlkan Ceylan[†], Jinwoo Kim[†]
arXiv 2026

P2. **Multi-view Relational Distillation for Spatial Reasoning with Vision-Language Models**

Kiet T. Nguyen, Hanbo Shim, Jinwoo Kim[†], Seunghoon Hong[†]
Under review

W2. **RelAgent: LLM Agents as Data Scientists for Relational Learning**

Xingyue Huang, Louis Tichelman, Jinwoo Kim, Krzysztof Olejniczak, İsmail İlkan Ceylan
ICML 2026 Workshop on Graph Foundation Models

C13. **Inverting Data Transformations via Diffusion Sampling**

Jinwoo Kim^{*}, Sékou-Oumar Kaba^{*}, Jiyun Park, Seunghoon Hong[†], Siamak Ravanbakhsh[†]
ICML 2026

P1. **Flow Map Language Models: One-step Language Modeling via Continuous Denoising**

Chanhyuk Lee, Jaehoon Yoo, Manan Agarwal, Sheel Shah, Jerry Huang, Aditi Raghunathan, Seunghoon Hong, Nicholas M. Boffi[†], Jinwoo Kim[†]
arXiv 2026

- C12. Flock: A Knowledge Graph Foundation Model via Learning on Random Walks**
Jinwoo Kim*, Xingyue Huang*, Krzysztof Olejniczak, Kyungbin Min, Michael Bronstein, Seunghoon Hong,
 Ismail Ilkan Ceylan
 ICLR 2026
- C11. Sequence Modeling with Spectral Mean Flows**
Jinwoo Kim, Max Beier, Petar Bevanda, Nayun Kim, Seunghoon Hong
 NeurIPS 2025
- C10. Revisiting Random Walks for Learning on Graphs**
Jinwoo Kim, Olga Zaghen*, Ayhan Suleymanzade*, Youngmin Ryou, Seunghoon Hong
 ICLR 2025
Spotlight Presentation (380/11672=3.26%)
 ELLIS Mobility Grant, ICML 2024 GRaM Workshop
- C9. 3D Denoisers are Good 2D Teachers: Molecular Pretraining via Denoising and Cross-Modal Distillation**
 Sungjun Cho, Dae-Woong Jeong, Sung Moon Ko, Jinwoo Kim, Sehui Han, Seunghoon Hong, Honglak Lee, Moontae Lee
 AAAI 2025
Oral Presentation
- C8. Simulation-Free Training of Neural ODEs on Paired Data**
 Semin Kim*, Jaehoon Yoo*, Jinwoo Kim, Yeonwoo Cha, Saehoon Kim, Seunghoon Hong
 NeurIPS 2024
- W1. Learning Symmetrization for Equivariance with Orbit Distance Minimization**
 Tien Dat Nguyen*, Jinwoo Kim*, Hongseok Yang, Seunghoon Hong
 NeurIPS 2023 NeurReps Workshop
- C7. Learning Probabilistic Symmetrization for Architecture Agnostic Equivariance**
Jinwoo Kim, Tien Dat Nguyen, Ayhan Suleymanzade, Hyeokjun An, Seunghoon Hong
 NeurIPS 2023
Spotlight Presentation (378/12345=3.06%)
- C6. Universal Few-shot Learning of Dense Prediction Tasks with Visual Token Matching**
 Donggyun Kim, Jinwoo Kim, Seongwoong Cho, Chong Luo, Seunghoon Hong
 ICLR 2023
Outstanding Paper Award (4/4955=0.08%)
 Silver Prize, Samsung Humantech Paper Award, 2023
- C5. Pure Transformers are Powerful Graph Learners**
Jinwoo Kim, Tien Dat Nguyen, Seonwoo Min, Sungjun Cho, Moontae Lee, Honglak Lee[†], Seunghoon Hong[†]
 NeurIPS 2022
- C4. Transformers meet Stochastic Block Models: Attention with Data-Adaptive Sparsity and Cost**
 Sungjun Cho, Seonwoo Min, Jinwoo Kim, Moontae Lee, Honglak Lee, Seunghoon Hong
 NeurIPS 2022
- C3. Equivariant Hypergraph Neural Networks**
Jinwoo Kim, Saeyoon Oh, Sungjun Cho, Seunghoon Hong
 ECCV 2022
- C2. Transformers Generalize DeepSets and Can be Extended to Graphs and Hypergraphs**
Jinwoo Kim, Saeyoon Oh, Seunghoon Hong
 NeurIPS 2021
 Qualcomm Innovation Fellowship Korea, 2021
- C1. SetVAE: Learning Hierarchical Composition for Generative Modeling of Set-Structured Data**
Jinwoo Kim*, Jaehoon Yoo*, Juho Lee, Seunghoon Hong
 CVPR 2021
- J1. Spontaneous Retinal Waves Can Generate Long-Range Horizontal Connectivity in Visual Cortex**
Jinwoo Kim*, Min Song*, Jaeson Jang, Se-Bum Paik
 The Journal of Neuroscience 40(34) 2020

Honors

Awards

Best Ph.D. Dissertation Award, KAIST College of Engineering	2026
Outstanding Researcher Award, KAIST-Mila Prefrontal AI Research Center	2024
ELLIS Mobility Grant [C10], ICML 2024 GRaM Workshop	2024
Outstanding Paper Award [C6], ICLR 2023	2023
Silver Prize, Samsung Humantech Paper Award [C6]	2023
Qualcomm Innovation Fellowship Korea [C2]	2022
KAIST Engineering Innovator Award (five recipients), KAIST College of Engineering	2020

Scholarships and Fellowships

Kwanjeong Education Foundation Scholarship	2022 – 2023
Korea National Science & Technology Scholarship	2018 – 2019
KAIST Alumni Fellowship	2017 – 2020
KAIST Presidential Fellowship	2016 – 2020
Hansung Scholarship for Gifted Students	2015 – 2016

Invited Talks

Flow Map Language Models: One-step Language Modeling via Continuous Denoising [P1]	
• Discrete Diffusion Reading Group (host: Subham Sahoo, Zhihan Yang)	04/2026
Sequence Modeling with Spectral Mean Flows [C11]	
• Ben-Gurion University of the Negev (BGU) (host: Ilan Naiman)	12/2025
Architecture-Agnostic Invariances for Deep Learning [C7, W1, C10, C12, C13]	
• Mila - Quebec AI Institute (host: Minsu Kim, Junyeob Baek)	07/2025
• KAIST AI899 Geometric DL (host: Sungsoo Ahn)	05/2025
• Mila - Quebec AI Institute (host: Siamak Ravanbakhsh, Sékou-Oumar Kaba)	12/2024
• KAIST-Mila Prefrontal AI Research Center (host: Sungjin Ahn)	11/2024
• Sungkyunkwan University (SKKU) (host: Chang Woo Myung)	08/2024
• Pohang University of Science and Technology (POSTECH) (host: Sungsoo Ahn)	11/2023
Universal Few-shot Learning of Dense Prediction Tasks with Visual Token Matching [C6]	
• KAIST–Samsung Electronics DS Division Exchange Meetup (host: Chulmoo Kang)	08/2023
Pure Transformers are Powerful Graph Learners [C5]	
• Microsoft USA (host: Nabiha Asghar)	01/2023
• NeurIPS 2022 at KAIST (host: Dongkwan Kim)	11/2022
• Learning on Graphs and Geometry Reading Group (LoGaG) (host: Hannes Stärk)	08/2022
Higher-order Transformers for Sets, Graphs, and Hypergraphs [C2]	
• Qualcomm Korea (host: Jaewon Choi)	01/2023
• KAIST AI Workshop 21/22 (host: Dongkwan Kim)	01/2022
• NeurIPS 2021 Social: ML in Korea (host: Jung-Woo Ha)	12/2021
Hierarchical Variational Autoencoders for Generative Modeling of Sets [C1]	
• Naver AI Author Meetup for CVPR 2021 (host: Jung-Woo Ha)	09/2021
• Korean Conference on Computer Vision 2021 (host: Jongwoo Lim)	09/2021
Retinal Waves and Prenatal Wiring of Primary Visual Cortex [J1]	
• Society for Neuroscience, Chicago, IL, US	10/2019

Academic Services

Workshop Area Chair: ICML 2026 GFM
Conference Reviewer: NeurIPS, ICLR, ICML, CVPR, ICCV, AISTATS, LoG, TAG-DS, IJCNN, ACCV
Journal Reviewer: TMLR, Neural Networks, IJCV

Teaching

Teaching Assistant, KAIST School of Computing

Undergraduate Research Program (URP)

2022, 2024

Introduction to Deep Learning (CS492I)

2021, 2022, 2023

Computer Vision (CS576)

2022, 2023

School of Computing Colloquium (CS966, CS986)

2021

Teaching Assistant, Samsung Electronics

Samsung Research AI Expert Program

2021 – 2024

Mentoring

Kiet T. Nguyen, M.S. student @ KAIST [P2]

2026 – present

Chanhyuk Lee, M.S. student @ KAIST [P1]

2025 – present

Kyungbin Min, B.S. student @ KAIST [C12] → M.S. student @ KAIST

2025 – present

Jiyeon Park, B.S. student @ KAIST [C13] → M.S. student @ KAIST

2024 – present

Nayun Kim, B.S. student @ KAIST [C11] → Intern @ EPFL LTS4

2024 – present

Ayhan Suleymanzade, B.S. student @ KAIST [C7, C10, P3] → Ph.D. student @ EPFL

2023 – present

Youngmin Ryou, B.S. student @ KAIST [C10] → on leave for mandatory military service

2023 – 2024

Nicole Shen, B.S. student @ MIT → Intern @ MIT LIDS

2024

Semin Kim, M.S. student @ KAIST [C8] → Ph.D. student @ KAIST

2023 – 2024

Olga Zaghen, M.S. student @ UniTrento [C10] → Ph.D. student @ UvA Amsterdam

2023

Tien Dat Nguyen, B.S. student @ KAIST [C5, C7, W1] → M.S. student @ UWaterloo

2021 – 2023

Daniel Sungho Jung, B.S. student @ Penn State → Ph.D. student @ SNU

2021

Saeyoon Oh, B.S. student @ KAIST [C2, C3] → Engineer @ FuriosaAI

2021

Projects

Korea National Research Foundation (NRF), Physical Reasoning in Language Models

2024 – 2025

Korea Ministry of Science and ICT, Visual Commonsense Reasoning

2021 – 2023

Korea National Research Foundation (NRF), Cooperative Robotic Intelligence

2021 – 2023

References

Seunghoon Hong, Associate Professor at KAIST

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İsmail İlkan Ceylan, Associate Professor at TU Wien

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